

Mark Li

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EXPERIENCE

Shure Inc.

Applications Development Engineer | Niles, IL

Jul. 2025 - Present

Applications Development Engineer Intern

Jun. 2024 - Mar. 2025

- Eliminated manual reporting and saved 5+ hours weekly for 60+ engineers by building an internal metrics platform aggregating CI, testing, and operational data across 7+ services
- Improved reliability of real-time device synchronization across distributed conferencing hardware deployments by developing production frontend systems for a large-scale enterprise audio management platform
- Enabled low-latency synchronization and monitoring across network-connected audio devices by integrating GraphQL-based real-time device state pipelines into the frontend
- Reduced engineering review bottlenecks by 50% building an internal hackathon tool that dynamically assigned PR reviewers based on repository ownership and reviewer load balancing

Northwestern University - McCormick School of Engineering

Teaching Assistant, Programming Languages (CS321) | Evanston, IL

Sep. 2023 - Mar. 2024

- Debugged and evaluated student implementations of interpreters and language features in Racket (PLAI), guiding a cohort of 70+ students through semantic and implementation errors

Explore (Startup)

Full-Stack Developer | Evanston, IL

Jun. 2023 - Sep. 2023

- Built a full-stack operations platform using Vue.js, Go, and PostgreSQL used by 80+ volunteer teams globally to coordinate volunteering opportunities
- Designed PostgreSQL schemas and optimized query patterns supporting scalable CRUD workflows and low-latency data access

PROJECTS

Verdra | Static analysis tool for cost-aware code optimization

- Built and deployed a static analysis platform using AST and control-flow analysis to identify cloud cost and performance inefficiencies, including redundant database calls, oversized imports, and over-provisioned serverless workloads
- Implemented automated rule evaluation pipelines capable of detecting 10+ optimization patterns across real-world codebases to reduce infrastructure waste and developer debugging overhead

RESEARCH EXPERIENCE

Northwestern University - McCormick School of Engineering

Embodied Systems Lab

Graduate Researcher

Mar. 2024 - Jun. 2025

- Built fault-tolerant execution systems on MSP430 microcontrollers capable of recovering from intermittent power loss and constrained-memory failures in embedded environments
- Engineered low-overhead embedded recovery mechanisms under strict energy and memory constraints to maintain persistent execution state across device interruptions
- Developed and led a workshop on tradeoffs in electronics and computer systems, illustrating how hardware constraints shape computation, storage, and energy use

Programming Languages Lab

Research Assistant

Jun. 2022 - Mar. 2024

- Evaluated probabilistic contract enforcement strategies across 13+ benchmarks to analyze performance-correctness tradeoffs in Racket

EDUCATION

Northwestern University

Master of Science in Computer Science

Jun. 2025

Bachelor of Science in Computer Science

Evanston, IL

Bachelor of Arts in Music

SKILLS

Programming Languages: Python, TypeScript, JavaScript, C/C++, Go, Java

Frontend: React, React Native, Angular

Backend: Node.js, GraphQL, FastAPI, REST APIs, PostgreSQL, MongoDB, AWS

Languages: English (Native), Mandarin (Professional), Spanish (Conversational)